UNIT-3

1. Network Layer

The Network Layer is responsible for **delivering data packets from the source computer to the destination computer across different networks**.

Main Functions of Network Layer

1. Logical Addressing
2. Routing
3. Forwarding
4. Packetizing
5. Internetworking
6. Congestion Control
7. Fragmentation and Reassembly

Important Network Layer Protocols

- IPv4
- IPv6
- ICMP
- IPsec
- OSPF
- RIP
- BGP

2. Network Layer Design Issues

Network Layer Design Issues are the important problems or decisions that must be considered while designing the Network Layer.

Main Design Issues

1. **Services Provided to the Transport Layer**
    - The Network Layer must provide services to the Transport Layer.
    - It should decide how packets are delivered.
    - Example: reliable or unreliable delivery.
2. **Routing**
    - Determines the **best path** for a packet from source to destination.
    - Example: choosing the shortest or least-congested path.
3. **Forwarding**
    - Once a route is selected, the router must decide **where to send the packet next**.
4. **Congestion Control**
    - When too many packets enter the network, congestion occurs.
    - The Network Layer should control traffic to reduce **packet loss and delay**.
5. **Quality of Service (QoS)**
    - Ensures that important applications receive suitable network performance.
    - Important parameters include:

- Bandwidth
- Delay
- Jitter
- Packet loss

6. **Internetworking**
   - Different types of networks must communicate with each other.
   - Example: LAN communicating with another LAN through the Internet.
7. **Addressing**
   - Each device must have a unique **logical address (IP address)**.
   - Example: 192.168.1.10
8. **Packet Fragmentation**
   - A large packet may need to be divided into smaller packets when the network cannot carry the original size.

3. Store and Forward Packet Switching

Store-and-Forward Packet Switching is a method in which a **router first receives and stores the complete packet**, checks it, and then **forwards it to the next router**.

Example

Suppose a packet has to travel:

Computer A → Router 1 → Router 2 → Computer B

The process is:

Step-1: Computer A Sends the packet to Router1

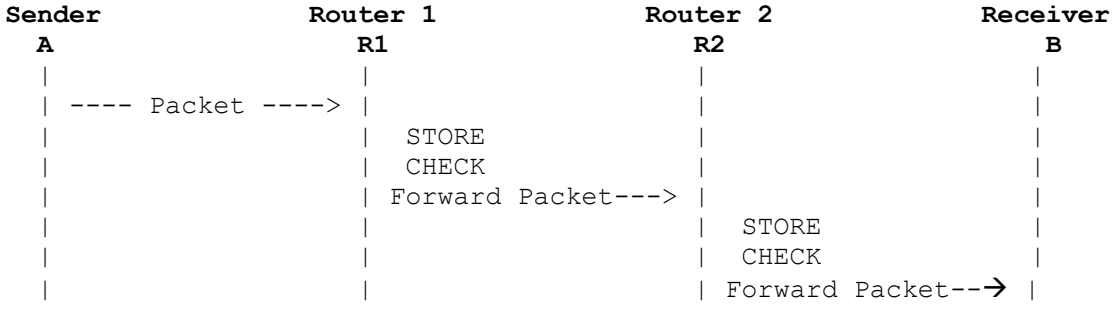
Step-2: Router1 stores the complete packet in memory

Step-3: Router1 checks the packet and determine where to send it

Step-4: Router1 forwards the packet to Router2

Step-5: Router2 again stores the packet and then forward it

Step-6: Finally the packet reaches Computer B



Representation of Store and Forward Packet Switching

Store → Router receives and stores the **entire packet**.

Forward → After receiving the complete packet, the router forwards it to the next router.

Advantages

- Helps detect damaged or corrupted packets before forwarding.
- Each router can make an appropriate forwarding decision.
- Useful for handling different network speeds.

Disadvantage

- Causes **delay**, because the router must receive and store the complete packet before forwarding it.

4. Services Provided to Transport Layer

The **Network Layer** is the third layer of the OSI model. It provides services to the **Transport Layer** above it. Its main responsibility is to provide a suitable method for transferring **packets from the source host to the destination host** across one or more networks.

The Network Layer should provide services that are **efficient, reliable where required, and independent of the internal details of the network**.

1. Connection-Oriented Service

In a connection-oriented service, a **logical connection is established before data transmission**.

The communication takes place in three stages:

```
Connection Establishment
        ↓
     Data Transfer
        ↓
 Connection Termination
```

Example: Virtual-circuit networks.

2. Connectionless Service

In a connectionless service, **no connection is established before sending data**.

Each packet is treated independently and may follow a different route.

```
Packet 1 → Route A
Packet 2 → Route B
Packet 3 → Route A
```

Example: IP provides a connectionless datagram service.

3. Reliable Delivery

The Network Layer may provide mechanisms to ensure that packets reach the destination correctly.

It can help handle problems such as:

- Packet loss
- Errors
- Network failures

However, in the Internet, **IP itself does not guarantee reliable delivery**; reliability is commonly provided by the Transport Layer, such as TCP.

4. Logical Addressing

The Network Layer provides **logical addresses**, such as IP addresses, to identify source and destination hosts.

Example:

```
Source IP       : 192.168.1.10
Destination IP  : 192.168.1.20
```

This allows packets to be delivered to the correct destination.

5. Routing

The Network Layer determines a suitable **path from the source to the destination**.

```
Source
  ↓
Router 1
  ↓
Router 2
  ↓
Router 3
  ↓
Destination
```

Routing protocols such as **RIP, OSPF, and BGP** help determine routes.

6. Forwarding

After selecting a route, routers perform **forwarding**.

Forwarding means moving a packet from an incoming interface to the appropriate **outgoing interface** toward its destination.

```
Incoming Packet
      ↓
    Router
      ↓
Outgoing Interface
      ↓
 Next Router
```

7. Quality of Service (QoS)

The Network Layer can support requirements for applications that need good network performance.

Important QoS parameters include:

- **Bandwidth**
- **Delay**
- **Jitter**
- **Packet loss**

For example, video conferencing requires low delay and low packet loss.

8. Congestion Control

When too many packets enter a network, **congestion** can occur.

The Network Layer can use congestion-management techniques to reduce:

- Packet loss
- Delay
- Network overload

9. Packet Sequencing

In some network services, packets may need to be delivered in the **correct order**.

For example:

```
Packet 1 → Packet 2 → Packet 3
```

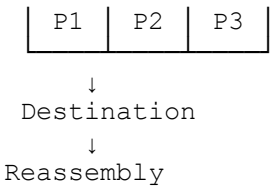
should reach the destination in the required sequence.

10. Fragmentation and Reassembly

Different networks may support different maximum packet sizes.

If a packet is too large, it may need to be **fragmented into smaller pieces**.

```
Large Packet
     ↓
 ┌─────┬─────┬─────┐
```



11. Internetworking

The Network Layer allows **different networks and network technologies to communicate** with each other.

For example:

```
LAN → Router → Internet → Router → LAN
```

This is an important function of the Internet Protocol.

12. Addressing and Delivery Across Networks

The Network Layer ensures that packets can travel across **multiple interconnected networks** using logical addressing and routing.

For example:

```
Computer A
    ↓
  LAN 1
    ↓
 Router
    ↓
 Internet
    ↓
 Router
    ↓
  LAN 2
    ↓
Computer B
```

5. Implementation of Connectionless Service

A **connectionless service** is a network service in which **no connection is established before data transmission**.

Each packet is treated as an **independent unit**, called a **datagram**. Every datagram contains the destination address and may travel through a different route.

Example: The Internet Protocol (**IP**) provides a connectionless datagram service.

Packet 1 ——————→ Router B ——————→ Destination

Packet 2—→ Router A ————————————→ Destination

Packet 3 ——————→ Router B ——————→ Destination

The packets do not need to follow the same path.

Implementation of Connectionless Service

he implementation can be understood in the following steps:

1. Packet Creation

The Transport Layer gives data to the Network Layer.

The Network Layer divides the data into **packets/datagrams**.

```
Transport Layer
       ↓
     Data
       ↓
Network Layer
       ↓
   Datagram
```

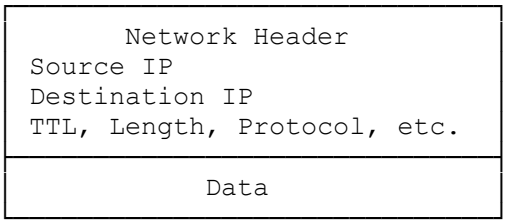
2. Adding Header

The Network Layer adds a header to each datagram.

The header contains important information such as:

- Source IP address
- Destination IP address
- Packet length
- Identification information
- Time-to-live (TTL)
- Protocol information

Example:



3. Forwarding the Datagram

The source sends the datagram to the first router.

The router examines the **destination IP address** and checks its forwarding/routing table.

```
Source
```

```
   ↓
Router 1
   ↓
Router 2
   ↓
Destination
```

4. Independent Routing

Each datagram is handled independently.

Therefore, different packets can take different paths.

```
Packet 1 → Router A → Router C → Destination
Packet 2 → Router B → Router D → Destination
Packet 3 → Router A → Router C → Destination
```

This is the main feature of connectionless service.

5. Store and Forward

A router normally:

1. Receives the complete packet.
2. Stores it temporarily.
3. Examines the destination address.
4. Determines the next hop.
5. Forwards the packet.

This is called **store-and-forward packet switching**.

6. No Connection Establishment

Unlike connection-oriented service, there is **no setup phase**.

```
Connectionless:

Send Packet 1
Send Packet 2
Send Packet 3
      ↓
No connection setup required
```

This makes connectionless communication relatively simple and flexible.

7. Handling Packets at the Destination

When packets reach the destination, the Network Layer passes them to the **Transport Layer**.

If packets arrive out of order, the Transport Layer may handle reordering when the transport protocol supports it.

For example, **TCP** can reorder segments and provide reliable delivery.

6. Implementation of Connection-Oriented Service

A **connection-oriented service** is a Network Layer service in which a **logical connection is established between the source and destination before data is transmitted**.

All packets belonging to the same connection generally follow the **same logical path**.

It is similar to making a **telephone call**: first the connection is established, then communication takes place, and finally the connection is terminated.

Three Phases of Connection-Oriented Service

The implementation consists of **three main phases**:

```
1. Connection Establishment
          ↓
2. Data Transfer
          ↓
3. Connection Termination
```

1. Connection Establishment

Before sending data, the source requests a connection with the destination.

The network selects a suitable route and establishes a **virtual circuit**.

```
Source              Router 1        Router 2          Destination
  |                    |               |                   |
  |--- Connection ---->|               |                   |
  |                    |--- Setup ---->|                   |
  |                    |               |--- Setup -------->|
  |                    |               |                   |
  |<------ Connection Established -------------------------|
```

After this process, the network knows the logical path between the source and destination.

Example

```
A → R1 → R2 → R3 → B
```

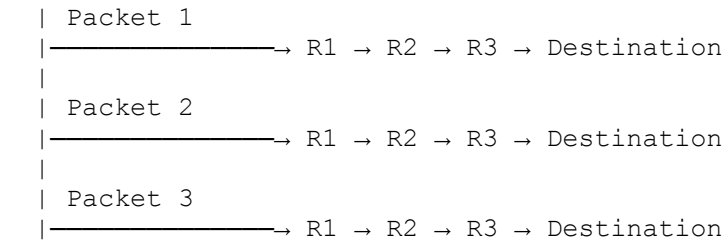
This path becomes the logical connection for the communication.

2. Data Transfer

After the connection is established, the source starts sending packets.

The packets belonging to the connection generally follow the **same established virtual circuit**.

```
Source
  |
```



Virtual Circuit Identifier

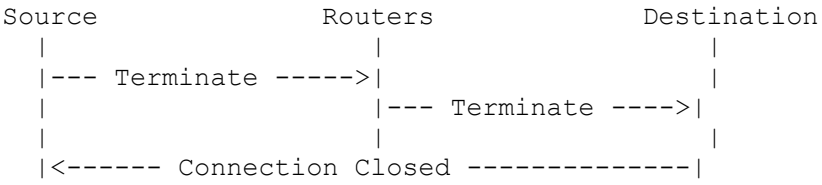
Each packet may contain a **virtual circuit identifier (VCI)** rather than carrying the complete destination information for routing at every hop.

The routers use the VCI to determine where to forward the packet.

3. Connection Termination

After all data has been transmitted, the source or destination requests that the connection be terminated.

The network removes the information associated with the virtual circuit.



The resources used by the connection can then be released.

7. Comparision of Virtual Circuit and Datagram Circuit

A **Virtual Circuit (VC)** and a **Datagram** are two approaches used for packet switching at the Network Layer.

Virtual Circuit:
First establish a path → then send all packets through that path.

Datagram:
Send each packet independently → each packet may take a different path.

Feature	Virtual Circuit	Datagram
Connection	Connection is established before data transfer	No connection establishment
Path	A logical path is established	No fixed path
Packet treatment	Packets belong to the same connection	Each packet is treated independently

Feature	Virtual Circuit	Datagram
Routing	Route is selected during connection setup	Route is determined for each packet
Packet header	Usually contains a short VC identifier	Contains the destination address
Router information	Routers maintain VC information	Routers generally do not maintain connection state
Path followed	Packets generally follow the same path	Packets may follow different paths
Packet order	Packets generally arrive in order	Packets may arrive out of order
Failure	Failure of the path can affect the connection	Packets can potentially be rerouted
Setup delay	Required	Not required
Overhead	Setup and connection-maintenance overhead	No connection setup overhead
Example	ATM, Frame Relay	IP datagram service

8. Routing Algorithms in Single Network

A **routing algorithm** is a method used by the **network layer** to decide the best path for sending packets from a **source node to a destination node**.

In a single network, routers communicate with each other and select a suitable route based on factors such as:

- Distance
- Cost
- Delay
- Number of hops
- Bandwidth
- Network congestion

Routing Algorithms:

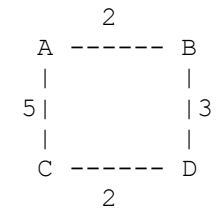
1.Shortest Path Routing:

The **shortest path algorithm** selects the path with the **lowest cost** between source and destination.

The cost may represent:

- Number of hops
- Distance
- Delay
- Link cost

Example:



Possible paths from A to D:

- A → B → D = 2 + 3 = **5**
- A → C → D = 5 + 2 = **7**

Therefore:

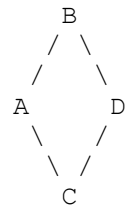
A → B → D is selected.

Common algorithm:

Dijkstra's algorithm is commonly used to find the shortest path.

2.Flooding:

In **flooding**, a router sends the received packet through **all outgoing links**, except the link from which it received the packet.



If A receives a packet, it forwards it to **B and C**.

The packet is eventually able to reach D through different paths.

Advantages

- Very reliable
- Does not require complete routing information
- Useful in emergency situations

Disadvantages

- Generates many duplicate packets
- Wastes bandwidth

- Can cause congestion

Control methods

To prevent unlimited flooding:

- Hop count can be used.
- Sequence numbers can be used.
- A packet can be forwarded only once.

3.Distance Vector Routing:

In **Distance Vector Routing**, each router maintains information about:

- Destination
- Distance/cost
- Next hop

Each router shares its routing information with its **neighboring routers**.

Example:

```
A ----- B ----- C
```

A initially knows only about B.

B tells A that:

B → C = 1 hop

Therefore A learns:

A → B → C = 2 hops

Routers tell their neighbors what they know about the network.

Advantages

- Simple
- Easy to implement
- Requires less information

Disadvantages

- Slow convergence
- Routing loops may occur
- Count-to-infinity problem may occur

Example protocol:

RIP (Routing Information Protocol) uses distance-vector principles.

4.Link State Routing:

In **Link State Routing**, each router learns the condition and cost of links in the network.

The router builds a **complete network map** and calculates the shortest path.

Steps:

- Discover neighboring routers.
- Measure the cost of each link.
- Create a link-state packet.
- Share the information with other routers.
- Build a complete network map.
- Calculate the shortest path.

```
Router
  |
  ↓
Find neighbors
  |
  ↓
Measure link cost
  |
  ↓
Share link-state information
  |
  ↓
Build network map
  |
  ↓
Find shortest path
```

Advantages

- Faster convergence
- Better knowledge of the network
- Less chance of routing loops

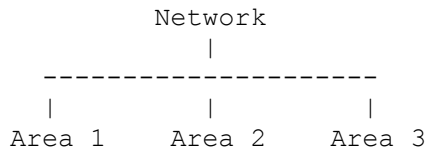
Disadvantages

- More complex
- Requires more memory and processing

5.Hierarchical Routing:

In a large network, maintaining information about every router becomes difficult.

Therefore, the network is divided into **areas or regions**.



Routers mainly maintain detailed information about their own area.

Advantages

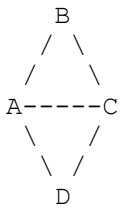
- Reduces routing information
- Saves memory
- Suitable for large networks
- Improves scalability

6.Broadcast Routing:

Broadcast routing is used when a packet needs to be delivered to **all nodes** in the network.

Example:

A router sends an important message to every router.



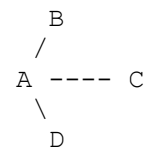
A broadcast packet is delivered to **B, C, and D**.

Uses:

- Network announcements
- Routing information
- Service discovery

7.Multicast Routing:

In **multicast routing**, a packet is sent to a **specific group of receivers**, not to everyone.



Suppose B and D belong to a multicast group.

The packet is sent only to:

B and D

It is not sent to C.

Advantage:

It saves bandwidth compared with sending separate packets to each receiver.

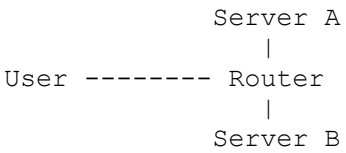
Applications:

- Video conferencing
- Live streaming
- Online classes
- Group communication

8.Anycast Routing:

In **anycast routing**, a packet is sent to **one member of a group**, usually the nearest or best available one.

Example:



If Server A and Server B provide the same service, the router selects the suitable server.

Simple meaning:

Anycast = Send the packet to one suitable server from a group of servers.

9. Traffic Management at Network Layer

Traffic management at the network layer means **controlling the flow of packets in a network** so that the network does not become overloaded or congested.

Traffic management = Managing network traffic to avoid congestion and ensure smooth packet delivery.

When too many packets enter the network at the same time, routers may become overloaded. This can cause:

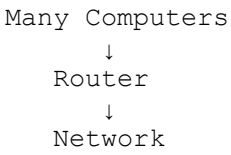
- Packet loss
- High delay
- Low throughput
- Network congestion

Need for Traffic Management

Traffic management is required to:

1. Prevent network congestion.
2. Reduce packet loss.
3. Control packet flow.
4. Improve network performance.
5. Maintain acceptable delay.
6. Provide better Quality of Service (QoS).

Example



If many computers send packets simultaneously, the router may receive more packets than it can process.

Therefore, **traffic must be controlled**.

Traffic Management Techniques:

1.Traffic Shaping:

Traffic shaping controls the rate at which packets are sent into the network.

Traffic shaping = Sending packets at a controlled and smooth rate.

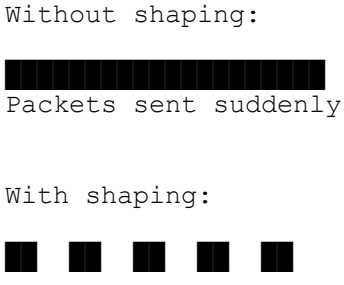
It prevents sudden bursts of packets from entering the network.

Example

Suppose an application generates:

100 packets suddenly

Traffic shaping may send them gradually instead of sending all 100 at once.



```
Packets sent smoothly
```

Advantages

- Reduces congestion
- Controls traffic bursts
- Improves network performance
- Provides smoother traffic flow

Traffic Shapping Algorithms:

1. **Leaky Bucket**
2. **Token Bucket**

2.Traffic Policing:

Traffic policing monitors traffic and checks whether it follows the allowed rate.

Traffic policing = Checking whether network traffic follows the specified limit.

If traffic exceeds the allowed limit, packets may be:

- Dropped
- Delayed
- Marked with lower priority

Example

Suppose a network allows:

10 Mbps

If a user sends:

15 Mbps

the extra traffic may be dropped or marked.

3. Packet Scheduling:

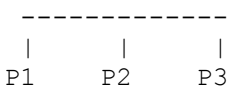
Packet scheduling is the process of deciding **which packet should be transmitted first** when multiple packets are waiting in a router.

Packet scheduling = Deciding the order in which packets are sent.

Example:

Suppose a router has three packets:

```
        Router
          |
```



The router uses a scheduling method to decide whether:

P1 → P2 → P3

or

P2 → P3 → P1

should be transmitted first.

Common scheduling methods:

- **FIFO (First In First Out):** First packet to arrive is sent first.
- **Priority Scheduling:** High-priority packets are sent first.
- **Round Robin:** Packets/classes are served one after another in turns.
- **Weighted Fair Queuing:** More important traffic can receive a larger share of bandwidth.

Example:

```
Voice packet     → High priority → Sent first
Video packet     → Medium priority
File packet      → Lower priority
```

Advantages:

- Reduces delay for important traffic.
- Provides better QoS.
- Uses bandwidth efficiently.

4.Admission Control:

Admission control decides whether a **new traffic flow or connection should be allowed into the network**.

Admission control = Check the network capacity before accepting new traffic.

Before accepting a new connection, the network checks resources such as:

- Bandwidth
- Buffer space
- Processing capacity
- Delay requirements

Example:

Suppose a network can support **10 video calls**.

```
Network capacity = 10 calls

Existing calls = 10

New call → ✗ Not admitted
```

If only 7 calls are active:

```
Existing calls = 7
Available capacity = 3

New call → ✓ Admitted
```

Advantages:

- Prevents network overload.
- Maintains QoS.
- Ensures sufficient resources for existing connections.

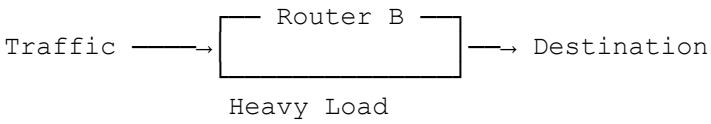
5.Load Balancing:

Load balancing distributes network traffic across **multiple paths, links, or routers**.

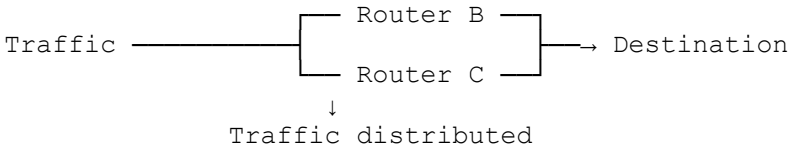
Load balancing = Distributing traffic so that one router or path does not become overloaded.

Example:

Without load balancing:



With load balancing:



For example, if 100 packets need to be sent:

- Path 1 → 50 packets
- Path 2 → 50 packets

Instead of sending all 100 packets through one path.

Advantages:

- Prevents overload.

- Improves throughput.
- Reduces congestion.
- Makes better use of available paths.
- Improves network reliability.

10. Internetworking

Internetworking means connecting **two or more different networks** so that devices in those networks can communicate with each other.

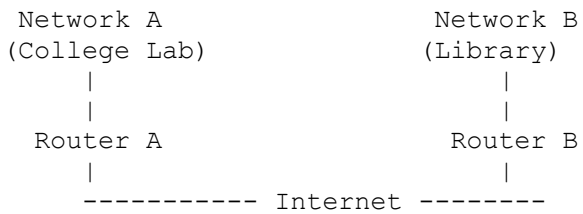
Internetworking = Connecting different networks to form a larger network.

For example, a college may have separate networks for:

- Computer Science Department
- Administration
- Library
- Wi-Fi

Internetworking allows these networks to communicate.

Example



The routers connect the different networks.

Need for Internetworking

Internetworking is required to:

1. Connect different networks.
2. Allow communication between networks.
3. Share resources.
4. Share information and services.
5. Provide communication over large geographical areas.
6. Connect different types of networks.

Overview of Internetworking:

Internetworking means connecting **two or more different networks** so that devices in those networks can communicate with each other.

Components of Internetworking:

The major components are:

1. Routers

A **router** connects different networks and forwards packets between them.

```
Network A → Router → Network B
```

2. Gateways

A **gateway** connects networks that may use different communication protocols or architectures.

3. Switches

A **switch** connects devices within the same local network.

4. Protocols

Protocols provide rules for communication.

Examples:

- IP
- TCP
- UDP
- ICMP

11. Connecting Heterogeneous Network

Connecting heterogeneous networks means connecting different types of networks so that they can communicate with each other.

A heterogeneous network **is a network made up of** different types of networks, technologies, devices, or protocols**.**

Heterogeneous = Different types

For example:

- Ethernet network
- Wi-Fi network
- Mobile network
- Fiber network
- Satellite network

A simple process is:

```
Network A
   ↓
Router/Gateway
   ↓
Common Network Protocol
   ↓
Router/Gateway
   ↓
Network B
```

Steps:

1. The source device creates a packet.
2. The packet is sent to the router.
3. The router examines the destination address.
4. The router selects an appropriate path.
5. The packet is forwarded through intermediate networks.
6. The destination network receives the packet.
7. The destination device receives the data.

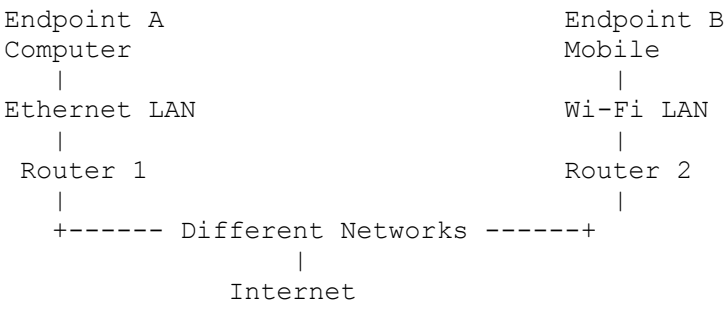
Connecting heterogeneous networks allows:

- Communication between different networks.
- Sharing of resources.
- Sharing of information.
- Access to common services.
- Communication between different devices.
- Internet connectivity.

12. Connecting endpoints across Heterogeneous Network

Connecting endpoints across heterogeneous networks means enabling two devices (endpoints) connected to **different types of networks** to communicate with each other.

Example

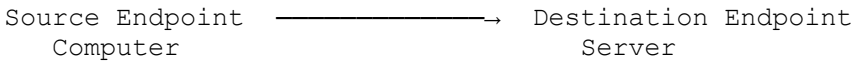


Here, the computer and mobile are connected through different network technologies, but they can still exchange data.

An **endpoint** is a device that sends or receives data.

Examples:

- Computer
- Mobile phone
- Server
- Printer
- IoT device



The **network layer** provides a common way to communicate across different networks.

The important components are:

1. IP Addressing

Each endpoint has an IP address that identifies it logically.

2. Routers

Routers forward packets from one network to another.

3. Routing

Routers determine a suitable path to the destination.

4. Network-Layer Protocol

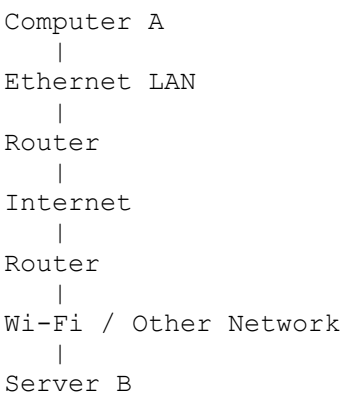
IP provides a common packet format and addressing mechanism across different underlying networks.

5. Gateways

Gateways may be required when protocol conversion or communication between substantially different network architectures is needed.

Communication Process

Suppose **Computer A** wants to communicate with **Server B**.



Step 1: Source Creates Data

Computer A creates data for Server B.

Step 2: Data is Encapsulated

The data is placed into a network-layer packet containing the **source and destination IP addresses**.

Step 3: Router Receives the Packet

The router examines the destination IP address.

Step 4: Routing

The router checks its routing information and selects the next hop.

Step 5: Crossing Different Networks

The packet can travel through different technologies such as:

Ethernet → Fiber → Wi-Fi → 5G

The underlying link-layer technology can change at each network.

Step 6: Destination Receives the Packet

Finally, the packet reaches Server B.

13. Internetwork Routing

Internetwork routing is the process of finding and selecting a path for packets to travel **from a source network to a destination network across multiple interconnected networks**.

Routing Across Multiple Networks

Suppose Computer A wants to send data to Server B.

```
Computer A
    ↓
Network A
    ↓
Router A
    ↓
Router B
    ↓
Router C
    ↓
Network B
    ↓
Server B
```

Steps:

1. Source sends the packet

Computer A creates a packet containing the destination IP address.

2. Router receives the packet

The first router examines the destination IP address.

3. Routing table is checked

The router checks its routing table to determine the next hop.

4. Packet is forwarded

The router sends the packet to the next router.

5. Intermediate routers repeat the process

Each router examines the destination and forwards the packet.

6. Destination is reached

Finally, the packet reaches the destination network and the destination device.

14. Supporting different packet sizes

Different networks may support **different maximum packet sizes**. The maximum amount of data that can be carried in one packet on a particular network is related to its **MTU (Maximum Transmission Unit)**.

Example

```
Network A          Network B          Network C
MTU = 1500         MTU = 1000         MTU = 1500

Computer → Router → Router → Server
```

A packet that fits in Network A may be **too large for Network B**.

MTU

MTU (Maximum Transmission Unit) is the largest IP packet size that a particular network link can carry without requiring fragmentation at that link.

For example:

```
Ethernet network → MTU = 1500 bytes
Another network  → MTU = 1000 bytes
```

If a 1500-byte packet needs to cross the 1000-byte network, special handling is required.

Fragmentation

Fragmentation is the process of dividing a large IP packet into **smaller fragments** so that each fragment can fit into the smaller MTU.

Example

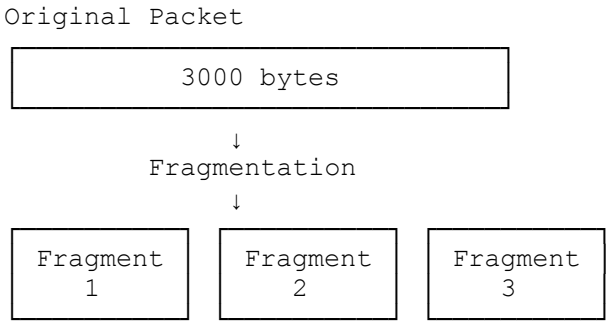
Suppose:

Original packet = 3000 bytes

The next network supports:

MTU = 1000 bytes

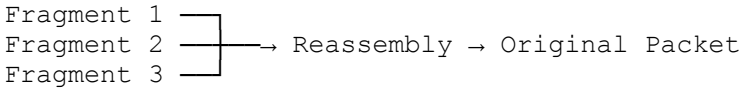
The large packet can be divided into smaller fragments.



Each fragment can travel through the smaller network.

Reassembly

At the destination, the fragments are combined to reconstruct the original packet.



15. Network Layer in Internet

The **Network Layer** is the **third layer of the OSI model**. It is responsible for delivering packets from a **source host to a destination host across one or more networks**.

The major functions of the network layer in the Internet are:

1. Logical addressing
2. Routing

3. Packet forwarding
4. Internetworking
5. Packet fragmentation/size handling
6. Error reporting and diagnostics
7. Support for different types of networks

IP – Internet Protocol

IP is the main network-layer protocol of the Internet.

Two important versions are:

- **IPv4**
- **IPv6**

IP provides **logical addressing** and enables packets to travel between different networks.

16. IPV4 Addressing and Packet Formats

Pv4 (Internet Protocol version 4) is a network-layer protocol used to identify devices and deliver packets across interconnected networks.

IPv4 uses a **32-bit address**.

An IPv4 address is **32 bits long**.

It is normally written as **four decimal numbers separated by dots**.

Example:

```
192.168.1.10
```

Each part is called an **octet** and represents **8 bits**.

```
192       .168       .1        .10
 8 bits    8 bits     8 bits     8 bits

       Total = 32 bits
```

An IPv4 address has two logical parts:

- **Network part** – identifies the network.
- **Host part** – identifies a device within that network.

```
Network part       Host part
192.168.1          .10
<------24 bits----> <8 bits>
```

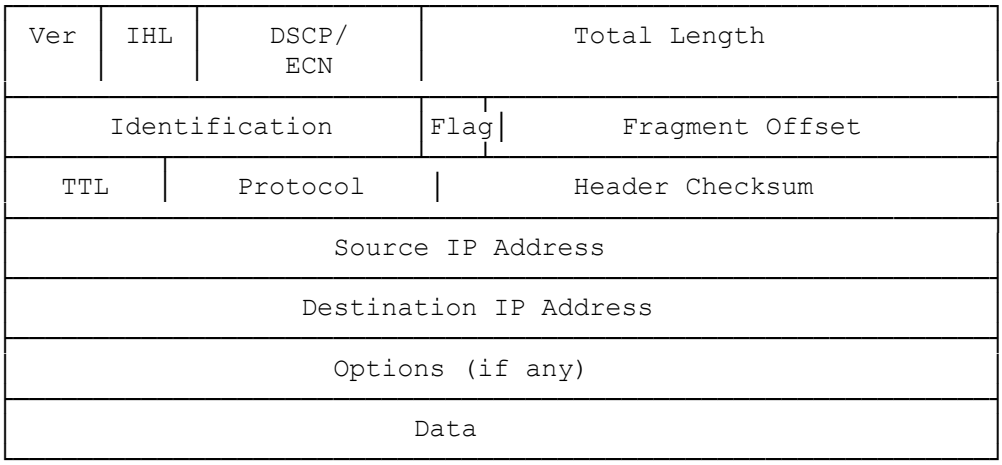
IPv4 Address Classes

Traditionally, IPv4 addresses were divided into five classes.

Class	First Octet	General Use
A	1–126	Very large networks
B	128–191	Medium-sized networks
C	192–223	Small networks
D	224–239	Multicast
E	240–255	Experimental

IPv4 Header Format

The IPv4 header has a **minimum size of 20 bytes** and can be larger when options are present.



Important IPv4 Header Fields

1. Version

Indicates the IP version.

For IPv4:

Version = 4

2. IHL

IHL = Internet Header Length

It specifies the length of the IPv4 header.

- Minimum = **20 bytes**
- Maximum = **60 bytes**

3. Total Length

Specifies the **total size of the IPv4 packet**, including:

Header + Data

Maximum IPv4 packet size:

65,535 bytes

4. Identification

Used to identify fragments belonging to the **same original packet**.

5. Flags

Contains control information related to fragmentation.

Important flags include:

- **DF – Don't Fragment**
- **MF – More Fragments**

6. Fragment Offset

Indicates the position of a fragment within the original packet.

It helps the destination **reassemble fragments correctly**.

7. TTL

TTL = Time To Live

It limits how long a packet can remain in the network.

Each router normally decreases the TTL by **1**.

If TTL reaches **0**, the packet is discarded.

Purpose:

Prevents packets from circulating forever because of routing loops.

8. Protocol

Identifies the **upper-layer protocol** carried inside the IP packet.

Examples:

- TCP → 6
- UDP → 17
- ICMP → 1

9. Header Checksum

Used to detect errors in the **IPv4 header**.

10. Source Address

Contains the IPv4 address of the sender.

Example:

```
192.168.1.10
```

11. Destination Address

Contains the IPv4 address of the intended receiver.

Example:

```
192.168.1.20
```

Routers use the destination address when making forwarding decisions.

12. Options

Optional information that may be included in the IPv4 header.

Because options are optional, the IPv4 header can be larger than 20 bytes.

16. IPV6 Addressing and Packet Formats

IPv6 (Internet Protocol version 6) is the newer version of the Internet Protocol developed to overcome the limitations of IPv4, especially the shortage of IPv4 addresses.

IPv6 Addressing

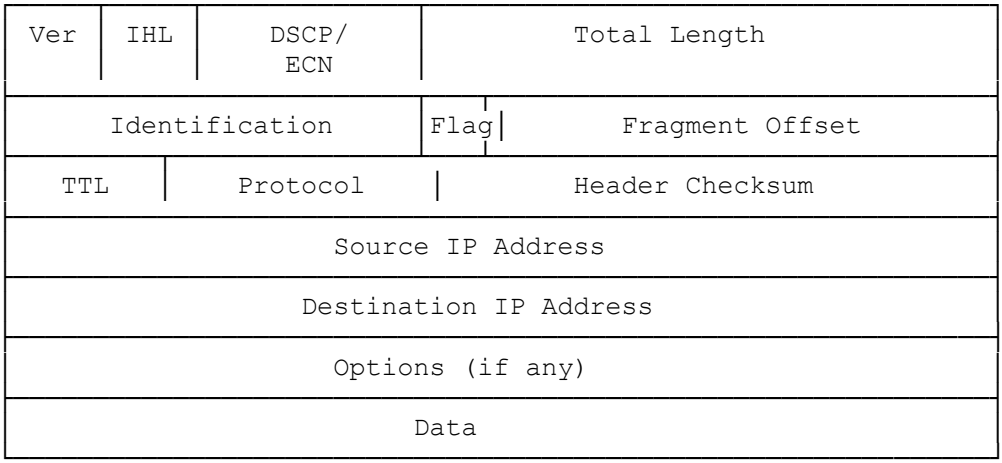
An IPv6 address is **128 bits long**.

It is written using **hexadecimal numbers**, divided into eight groups.

Example:

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

IPv6 Header Format



Important IPv6 Header Fields

1. Version

Indicates the IP version.

For IPv6:

Version = 6

2. IHL

IHL = Internet Header Length

It specifies the length of the IPv6 header.

- Minimum = **20 bytes**
- Maximum = **40 bytes**

3. Total Length

Specifies the **total size of the IPv6 packet**, including:

Header + Data

4. Identification

Used to identify fragments belonging to the **same original packet**.

5. Flags

Contains control information related to fragmentation.

Important flags include:

- **DF – Don't Fragment**
- **MF – More Fragments**

6. Fragment Offset

Indicates the position of a fragment within the original packet.

It helps the destination **reassemble fragments correctly**.

7. TTL

TTL = Time To Live

It limits how long a packet can remain in the network.

Each router normally decreases the TTL by **1**.

If TTL reaches **0**, the packet is discarded.

Purpose:

Prevents packets from circulating forever because of routing loops.

8. Protocol

Identifies the **upper-layer protocol** carried inside the IP packet.

Examples:

- TCP → 6
- UDP → 17
- ICMP → 1

9. Header Checksum

Used to detect errors in the **IPv6 header**.

10. Source Address

Contains the IPv6 address of the sender.

11. Destination Address

Contains the IPv6 address of the intended receiver.

12. Options

Optional information that may be included in the IPv6 header.

16. Internet Control Protocols

Internet Control Protocols are protocols used in the Internet to **send control, error, diagnostic, and management information**.

They do not mainly carry user data like web pages or files. Instead, they help the network **report problems, check connectivity, and manage IP communication**.

CMP

IGMP

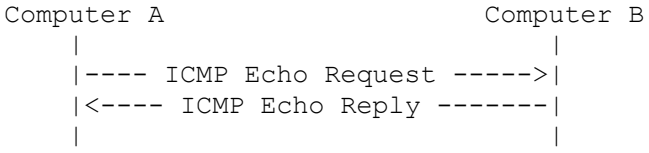
ICMPv6

Internet Control Message Protocol(ICMP)

ICMP is used to **send error messages and control information** in an IP network.

It helps us know whether a destination is reachable and reports problems when packets cannot be delivered.

Example: Ping



If Computer B replies, Computer A knows that B is reachable.

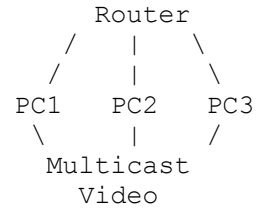
Main Functions

- Reports network errors
- Checks connectivity
- Helps troubleshoot networks
- Reports **Destination Unreachable**
- Reports **Time Exceeded**
- Supports tools such as **Ping** and **Traceroute**

Internet Group Management Protocol(IGMP)

IGMP is used to manage multicast group membership in IPv4 networks. Multicast means sending the same data to **a selected group of receivers**.

For example, a live video can be sent to several users who joined the same multicast group.



IGMP tells the router **which devices want to receive multicast traffic**.

Main Functions

- Join a multicast group
- Leave a multicast group
- Maintain multicast group membership
- Help routers manage multicast traffic

ICMPv6 – Internet Control Message Protocol Version 6

ICMPv6 stands for **Internet Control Message Protocol Version 6**.

It is a **Network Layer control protocol used with IPv6** to report errors, perform network diagnostics, and support important IPv6 functions.

Main Functions of ICMPv6

1. Error Reporting

ICMPv6 reports problems with IPv6 packets.

Examples:

- **Destination Unreachable**
- **Packet Too Big**
- **Time Exceeded**
- **Parameter Problem**

2. Connectivity Testing

ICMPv6 provides:

Echo Request → Echo Reply

```
Computer A                    Computer B
    |                             |
    |--- Echo Request ----------->|
    |<-- Echo Reply ---------------|
    |                             |
```

This is used by **ping** to check connectivity.

3. Neighbor Discovery

One of the most important uses of ICMPv6 is **Neighbor Discovery Protocol (NDP)**.

It helps IPv6 devices:

- Discover neighboring devices
- Find routers

- Determine link-layer addresses
- Check whether a neighbor is reachable
- Perform address configuration-related functions

17. Label Switching and MPLS

Label Switching is a technique in which routers use a **short label** to forward packets instead of examining the complete destination IP address at every router.

MPLS (Multiprotocol Label Switching) is the most common technology that uses label switching.

In normal IP routing:

```
Packet → Check Destination IP → Routing Table → Next Hop
```

In label switching:

```
Packet → Check Label → Label Table → Next Hop
```

MPLS = Multiprotocol Label Switching

MPLS is a networking technology that forwards packets using **labels**.

It can carry different types of network traffic, such as:

- IP packets
- Voice traffic
- Video traffic
- VPN traffic

The term **multiprotocol** means MPLS can work with different network-layer protocols and is not limited to one type of traffic.

18. OSPF and BGP Routing Protocols

Both **OSPF** and **BGP** are routing protocols used to find the best path for sending packets, but they are used in different types of networks.

1. OSPF – Open Shortest Path First

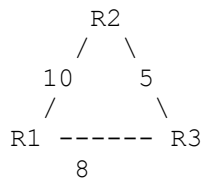
OSPF is an **Interior Gateway Protocol (IGP)** used for routing **inside one organization or Autonomous System (AS)**.

Working of OSPF

1. Routers discover their neighboring routers.
2. Each router collects information about the network links.
3. This information is shared with other OSPF routers.

4. Each router creates a **Link-State Database (LSDB)**.
5. OSPF uses **Dijkstra's Shortest Path First (SPF) algorithm**.
6. The best paths are placed in the routing table.

Example:



OSPF compares the **cost** of different paths and selects the path with the lowest total cost.

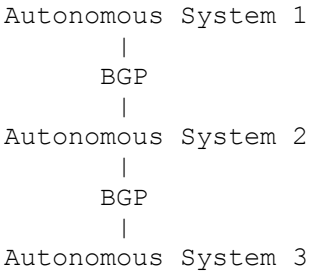
Important features of OSPF

- Link-state routing protocol
- Uses **Dijkstra's algorithm**
- Uses **cost** as its metric
- Supports hierarchical routing using **Areas**
- **Area 0** is the backbone area
- Supports fast convergence
- Supports IPv4 and IPv6
- Commonly used inside organizations

2. BGP – Border Gateway Protocol

BGP is an **Exterior Gateway Protocol (EGP)** used mainly for routing **between different Autonomous Systems (AS)**.

Think of the Internet as many different networks operated by different organizations. BGP helps these networks decide how traffic should travel **from one network to another**.



Working of BGP

1. BGP routers establish a connection with neighboring BGP routers.
2. They exchange information about reachable networks.
3. BGP receives multiple possible routes.
4. It applies its **path-selection rules**.
5. It selects the preferred route.
6. The selected route is added to the routing table.

Important features of BGP

- Path-vector routing protocol
- Used between Autonomous Systems
- Main routing protocol of the Internet
- Uses **TCP port 179**
- Supports policy-based routing
- Handles very large networks
- Uses **AS Path** to help select routes and prevent routing loops